AN ALGEBRAIC–VARIATIONAL FRAMEWORK FOR THE AFFINE PLANK PROBLEM

Abstract. Bang's affine (relative–width) plank conjecture asserts that if a convex body $K \subset \mathbb{R}^d$ is covered by finitely many planks, then the sum of the widths of the planks, each measured relative to the width of K in the corresponding direction, is at least 1. The conjecture is sharp, is known for centrally symmetric bodies by a theorem of Ball, and is otherwise open—open even in the plane and even for three planks. The best current lower bound for the total relative width of a covering of a general body is $2/(1+\sqrt{d})$, due to Bakaev and Yehudayoff. We isolate a methodological gap: the critical–point and Euler–Jacobi vanishing technique that Martínez and Ortega–Moreno recently used to settle the real linear polarization problem has not been brought to bear on planks. This paper sets up a precise framework for doing so. We recall the engine in the polarization setting, record the elementary dictionary between relative widths and gauge functionals, observe that in the centrally symmetric (Hilbertian) case the relative–width statement is exactly of polarization type and is therefore already accessible to the engine, and formulate the three analytic obstructions—reality of the critical locus, a Bernstein–Kushnirenko count of that locus, and a toric Euler–Jacobi vanishing identity—as explicit conjectures whose resolution would transport the method to the non–symmetric case. We are careful throughout to separate what is proved from what is proposed.

1. Introduction

1.1. **The plank problem and its affine refinement.** A *plank* of width $w \geq 0$ in $\mathbb{R}^d$ is the closed region between two parallel hyperplanes at Euclidean distance w; equivalently, for a unit vector $u \in S^{d-1}$ and reals $a \leq b$ with $b - a = w$,

$$P = \{x \in \mathbb{R}^d : \ a \leq \langle x, u\rangle \leq b\}.$$

For a convex body $C \subset \mathbb{R}^d$ (a compact convex set with nonempty interior), the *width of C in the direction u* is

$$\mathrm{w}(C; u) = \max_{x \in C}\langle x, u\rangle - \min_{x \in C}\langle x, u\rangle,$$

and the *minimal width* of C is $\mathrm{w}(C) = \min_{u \in S^{d-1}} \mathrm{w}(C; u)$.

In 1932 Tarski asked whether a convex body covered by planks must have the sum of the plank widths at least $\mathrm{w}(C)$; he settled only the disk [12]. Bang proved Tarski's conjecture in general [4]: if $C \subseteq P_1 \cup \cdots \cup P_n$ then $\sum_i \mathrm{w}(P_i) \geq \mathrm{w}(C)$. At the end of [4] Bang proposed a stronger, affinely invariant version, introducing the term *relative width*.

Definition 1.1. Let P be a plank with unit normal u and width $\mathrm{w}(P)$. The *relative width* of P with respect to a convex body C is

$$\mathrm{rw}_C(P) \ := \ \frac{\mathrm{w}(P)}{\mathrm{w}(C; u)}\,.$$

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Conjecture 1.2 (Bang's affine plank conjecture [4])**.** If a convex body $C \subset \mathbb{R}^d$ $(d \geq 2)$ is covered by planks $P_1, \ldots, P_n$, then

$$\sum_{i=1}^{n} \mathrm{rw}_C(P_i) \;\geq\; 1.$$

The quantity $\sum_i \mathrm{rw}_C(P_i)$ is invariant under invertible affine maps of $\mathbb{R}^d$, which is the sense in which Conjecture 1.2 refines Tarski's absolute–width problem. The bound is sharp: d parallel slabs of equal relative width $1/d$ stacked across C already attain equality in many configurations.

1.2. **What is known.** We summarize the state of the art; precise quotations and a verification log are collected in the companion notes accompanying this manuscript.

- **Symmetric case (solved).** Ball [2] proved Conjecture 1.2 for centrally symmetric bodies. In functional–analytic form: if the unit ball of a Banach space is covered by a countable family of planks, the sum of their half–widths is at least 1; dually, for a symmetric convex body C and n hyperplanes there is a translate of a copy of C scaled by at least $1/(n+1)$ lying inside C and avoiding all the hyperplanes. The proof extends Bang's Lemma and turns on the optimization of a suitable quadratic form over the sign vectors $\{-1, 1\}^n$ of a symmetric matrix.
- **General case (open).** For arbitrary convex bodies Conjecture 1.2 is open. It is open already in the plane, and is not known even for three planks; Bang himself verified only the case of two planks [4, 9, 13, 1]. The known lower bounds for the total relative width of a covering of a general body in $\mathbb{R}^d$ have improved from $1/d$ (John's theorem together with Bang's theorem; see [6, 7]) to $2/(1+d)$ [7] and, most recently, to $2/(1+\sqrt{d})$ [1], the current record. Chambers and Mouille [7] also prove the unconditional bound $\sum_i \mathrm{rw}_C(P_i) \geq 1/\dim \pi_V(C)$, where V is the span of the plank normals and π_V is orthogonal projection onto V, and they prove the sharp bound 1 *conditionally*, under the hypothesis that C minus the union of the first m planks remains convex for every m.

1.3. **The polarization connection and a methodological gap.** The plank problem is tightly linked to lower bounds for products of linear functionals. For a Banach space X, the nth linear polarization constant $c_n(X)$ is the least C with $\|f_1\| \cdots \|f_n\| \leq C\, \|f_1 \cdots f_n\|$ for all $f_1, \ldots, f_n \in X^*$; Ball's symmetric plank theorem yields the general real bound $c_n(X) \leq n^n$ [5, 11]. Over a real Hilbert space the sharp conjecture $c_n \leq n^{n/2}$ stood for decades and was recently established. Martínez and Ortega–Moreno [10] solved the real polarization problem by a variational analysis of the critical locus of a product of linear forms on the sphere, combined with the classical *Euler–Jacobi vanishing theorem*. Galicer, Ortega–Moreno and Pinasco [8] extended the argument to a weighted "strong" inequality and, as a corollary, to a centred plank theorem in Hilbert space.

These developments introduce into the plank/polarization circle a genuinely algebraic tool—counting the solutions of a polynomial critical system and forcing a good one through a vanishing identity. To the best of our knowledge, *this tool has not been applied to the affine plank problem.* The purpose of the present paper is to set up, precisely and honestly, a framework that would do so, and to delineate exactly which analytic facts must be established for the transfer to succeed.

1.4. **Status of results.** In keeping with the affine, sharp, and still–open nature of Conjecture 1.2, we are explicit about what this manuscript does and does not prove.

- Section 2 *recalls*, with attribution, the critical–point/Euler–Jacobi engine of [10, 8]. Nothing there is claimed as new.

- Section 3 proves elementary, self–contained facts: the affine invariance of the total relative width (Proposition 3.1), the gauge reformulation (Proposition 3.2), and the observation that in the centrally symmetric case the relevant statement is of polarization type and hence already within reach of the engine (Proposition 3.3). These are correct and complete.
- Section 4 states the three transfer obstructions as *conjectures and problems*. They are *not* theorems, are clearly labeled as such, and are never used as if proved.

No new lower bound for the general case, and in particular no improvement of $2/(1+\sqrt{d})$, is claimed here. The contribution of this paper is a framework and a programme, together with the elementary results that make the programme precise.

2. The critical–point/Euler–Jacobi engine, recalled

We recall the mechanism of [10] in the streamlined form of [8]. Nothing in this section is original; we include it because the framework of Section 4 is a structural analogy with it.

Let $v_1,\dots,v_n$ be a basis of $\mathbb{R}^n$ and let $k_1,\dots,k_n\in\mathbb{N}$, $s=\sum_j k_j$, $\alpha_j=k_j/s$. Consider

$$P(x)=\prod_{j=1}^{n}\langle v_j,x\rangle^{k_j},\qquad x\in S^{n-1},$$

and write $y_j=\langle v_j,x\rangle$. With $w_1,\dots,w_n$ the dual basis and $z_j=\langle w_j,x\rangle$ one has $x=\sum_j z_jv_j$ and $z=Ay$, where $A=G^{-1}$ and $G=(\langle v_i,v_j\rangle)_{i,j}$ is the Gram matrix.

Lemma 2.1 (Critical system; [10, 8])**.** *At a critical point of* $\log|P|$ *on the sphere with* $P\neq 0$ *one has* $\sum_j k_jv_j/y_j=s\,x$ *and* $y_jz_j=\alpha_j$. *Equivalently, the critical points are the solutions of the quadratic system* $h(y)=0$, *where*

$$h_j(y)=y_j(Ay)_j-\alpha_j,\qquad j=1,\dots,n.$$

Lemma 2.2 (Reality and count; [8, Lem. 3–4], [10])**.** *Every solution* $y\in\mathbb{C}^n$ *of* $h(y)=0$ *is real; in each open sign orthant of* $\mathbb{R}^n$ *there is exactly one solution; hence* h *has exactly* 2^n *solutions, all real and simple. At a solution,* $\mathrm{J}\,h(y)=\operatorname{diag}(y)\big(A+\operatorname{diag}(\alpha_j/y_j^2)\big)$, *which is invertible.*

Theorem 2.3 (Polarization via Euler–Jacobi; [10, 8])**.** *With the notation above there is a unit vector* $u\in S^{n-1}$ *with* $\sum_j k_j^2/\langle v_j,u\rangle^2\le s^2$, *equivalently* $\sum_j \alpha_j^2/\langle v_j,u\rangle^2\le 1$.

Sketch, after [8]. The system $h=0$ is a zero–dimensional complete intersection of n quadrics with $2^n=\prod_j\deg h_j$ simple solutions (Lemma 2.2), so Bézout's bound is attained and the Euler–Jacobi vanishing theorem applies: for every polynomial g with $\deg g\le n-1$,

$$\sum_{h(y)=0}\frac{g(y)}{\det \mathrm{J}\,h(y)}=0.$$

Choosing $g(y)=Q(y)\big(s^2-\sum_j k_j^2/y_j^2\big)$ with $Q(y)=\prod_j y_j$, and using $\det\mathrm{J}\,h(y)=Q(y)\det\big(A+\operatorname{diag}(\alpha_j/y_j^2)\big)$ with the second factor positive at every solution, the identity becomes a sum of the quantities $s^2-\sum_j k_j^2/y_j^2$ weighted by positive numbers. Hence some solution has $\sum_j k_j^2/y_j^2\le s^2$; its normalization is the desired u. Rational and real weights follow by approximation; see [8]. □

Remark 2.4. Theorem 2.3 rests on three pillars: *(P1) reality* of the critical locus, *(P2) an exact count* of that locus matching the product of the degrees, and *(P3) a vanishing identity* forcing a witness with the desired sign. The metric data enter only through the positive–definite matrix $A=G^{-1}$. The framework of Section 4 asks what survives when the round sphere is replaced by the boundary of a general convex body and A by the second–order data of that body.

3. Elementary reformulations

This section is self–contained and its results are proved in full.

3.1. Affine invariance.

Proposition 3.1. *Let $T : \mathbb{R}^d \to \mathbb{R}^d$ be an invertible affine map. If planks $P_1, \dots, P_n$ cover a convex body C, then $T(P_1), \dots, T(P_n)$ are planks covering $T(C)$ and*

$$\sum_{i=1}^{n} \mathrm{rw}_{T(C)}(T(P_i)) = \sum_{i=1}^{n} \mathrm{rw}_C(P_i).$$

Proof. Write $T(x) = Lx + b$ with L invertible. The image of a plank with normal u and width w is the region between the images of its two boundary hyperplanes; these remain parallel, so $T(P_i)$ is a plank, with some normal u_i' and width w_i'. For any direction, affine images of the two supporting hyperplanes of C that realize $\mathrm{w}(C; u)$ are supporting hyperplanes of $T(C)$ parallel to the images of the bounding hyperplanes of P_i. Both $\mathrm{w}(T(P_i))$ and $\mathrm{w}(T(C); u_i')$ are the distances between the corresponding pairs of parallel hyperplanes, measured along the same transversal; their ratio is therefore unchanged because it is computed within a single one–dimensional quotient $\mathbb{R}^d/$(common hyperplane direction) on which T acts by a single affine isomorphism. Hence $\mathrm{rw}_{T(C)}(T(P_i)) = \mathrm{rw}_C(P_i)$ for each i, and the covering $T(C) \subseteq \bigcup_i T(P_i)$ is immediate from $C \subseteq \bigcup_i P_i$. □

Proposition 3.1 is the reason Conjecture 1.2 is called *affine*: one may normalize C by any affine map (for example into John position) without changing the quantity under study.

3.2. Gauge reformulation.

For a convex body C with 0 in its interior, let $\|x\|_C = \inf\{t > 0 : x \in tC\}$ be its gauge (Minkowski functional) and $h_C(u) = \max_{x \in C}\langle x, u\rangle$ its support function. For a plank P with unit normal u given by $a \le \langle \cdot, u\rangle \le b$, one has $\mathrm{w}(P) = b - a$ and $\mathrm{w}(C; u) = h_C(u) + h_C(-u)$.

Proposition 3.2. *With C, P_i (normals u_i, levels $a_i \le b_i$) as above,*

$$\mathrm{rw}_C(P_i) = \frac{b_i - a_i}{h_C(u_i) + h_C(-u_i)}.$$

In particular, if C is centrally symmetric about 0 then $h_C(u) + h_C(-u) = 2h_C(u) = 2\|u\|_{C^\circ}$, where C° is the polar body, and $\mathrm{rw}_C(P_i) = (b_i - a_i)/\big(2\|u_i\|_{C^\circ}\big)$.

Proof. The first identity is the definition together with $\mathrm{w}(C; u) = h_C(u) + h_C(-u)$. If $C = -C$ then $h_C(-u) = h_C(u)$, and $h_C = \|\cdot\|_{C^\circ}$ by the definition of the polar body $C^\circ = \{u : \langle x, u\rangle \le 1 \ \forall x \in C\}$. □

3.3. The symmetric case is of polarization type.

The next proposition records that, for symmetric bodies, the relative–width statement is exactly the kind of inequality the engine of Section 2 produces. It is a reading of known results, included to mark the boundary of what is already accessible.

Proposition 3.3. *Let H be a real Hilbert space (so $C = B_H$ is the symmetric Euclidean ball). A family of planks with unit normals u_i and half–widths $t_i \ge 0$ fails to cover B_H as soon as $\sum_i t_i < 1$. Equivalently, in the dual formulation of [2, 8]: given unit vectors $v_1, \dots, v_n \in H$ and weights $p_i > 0$ with $\sum_i p_i = 1$, there is a unit vector u with $|\langle v_i, u\rangle| \ge p_i$ for all i; in particular the planks $\{x : |\langle v_i, x\rangle| < p_i\}$ do not cover the unit sphere.*

Proof. This is the centred Hilbert plank statement. The displayed dual form is [8, Cor. 6], itself a consequence of Theorem 2.3 via the weighted arithmetic–geometric mean inequality; the covering statement is the contrapositive. The symmetric body case of Conjecture 1.2 in full generality (countable coverings, sharp constant) is Ball's theorem [2]. □

Remark 3.4. Proposition 3.3 delineates the frontier precisely: for symmetric C the engine already applies (through the Gram matrix $A = G^{-1}$ of the normals, exactly as in Section 2). The open content of Conjecture 1.2 is entirely in the *non–symmetric* case, where $h_C(u)+h_C(-u)$ is genuinely asymmetric and no inner–product structure is given.

4. A programme for the non–symmetric case

We now state, as conjectures and problems, the analytic facts whose proof would transport the engine of Section 2 to general bodies. We stress that this section contains *no theorems*: the statements below are open and are offered as a research programme.

4.1. **The proposed critical system.** Fix a smooth, strictly convex body C with 0 in its interior and a covering by planks with normals $u_1, \ldots, u_n$. Guided by Section 2, one seeks an objective Φ on ∂C (or on a suitable gauge sphere) whose critical points encode the total relative width, with the role of the Gram matrix played by the second fundamental form / support–function Hessian of C. Making this precise is the first task.

Problem 4.1. Construct an objective functional Φ, polynomial or rational in suitable coordinates adapted to ∂C, whose stationarity conditions on ∂C form a zero–dimensional polynomial system $H_C(y) = 0$ encoding the quantities $\mathrm{rw}_C(P_i)$, and which reduces to the system h of Lemma 2.1 when C is an ellipsoid.

4.2. **The three obstructions.** Assume Problem 4.1 is solved, producing a system H_C with second–order data encoded by a positive–definite matrix field A_C (the inverse support–Hessian).

Conjecture 4.2 (P1: reality)**.** For smooth, strictly convex C every complex solution of $H_C(y) = 0$ is real.

The motivation is that the reality argument of [8, Lem. 3] uses only positive–definiteness of A through an imaginary–part energy identity; for a smooth strictly convex body the support–Hessian is positive definite, so the argument is expected to persist, possibly after a limiting passage to handle bodies that are not strictly convex (such as polytopes).

Conjecture 4.3 (P2: BKK count)**.** The system $H_C(y) = 0$ is a zero-dimensional complete intersection. Its number of solutions equals the Bernstein–Kushnirenko (BKK) mixed volume of the Newton polytopes of its components; all solutions are simple; and they are in bijection with the normal-fan cells of C cut out by the plank directions.

For the round sphere the Newton polytopes are those of the quadrics h_j and the BKK number specializes to the Bézout number 2^n of Lemma 2.2; Conjecture 4.3 asks for the analogue when the body is not an ellipsoid, where the mixed volume is in general smaller and structured.

Conjecture 4.4 (P3: toric Euler–Jacobi vanishing)**.** There is a test functional g_C, of degree below the relevant BKK threshold, for which the toric (global residue) form of the Euler–Jacobi vanishing theorem yields

$$\sum_{H_C(y)=0} \frac{g_C(y)}{\det \mathrm{J}\, H_C(y)} = 0,$$

and g_C can be chosen so that this identity forces the existence of a critical point witnessing $\sum_i \mathrm{rw}_C(P_i) \geq 1$.

Intended architecture (conditional, not a theorem). *If* Problem 4.1 is solved and Conjectures 4.2–4.4 hold for a class $\mathcal{K}$ of bodies, *then* Conjecture 1.2 would follow for every $C \in \mathcal{K}$. We state this only to display the intended logical architecture; it is contingent on three open conjectures and is deliberately *not* formulated as a theorem.

4.3. **Test classes and falsifiers.** The programme is to be validated or refuted on the smallest non–trivial classes before any attempt at generality.

Problem 4.5 (Symmetric recovery)**.** Carry out Problem 4.1 and Conjectures 4.2–4.4 for centrally symmetric C, thereby *reproving* Ball's theorem [2] by the critical–point/Euler–Jacobi method. (By Proposition 3.3 the required inputs are already available; the content is to phrase Ball's quadratic optimization as a special case of the engine.)

Problem 4.6 (Simplices)**.** Establish the sharp bound $\sum_i \mathrm{rw}_C(P_i) \geq 1$ for the simplex, the most tractable non–symmetric body. If a reduction of Conjecture 1.2 to simplices in the literature is correct, a positive answer here would settle the conjecture in a structured case for the first time by these methods.

Problem 4.7 (Planar three–plank case)**.** Decide Conjecture 1.2 for $d = 2$ and $n = 3$, where the critical system is of small size, by the present method. This case is currently open [13, 9].

A negative outcome on Problem 4.5 (the engine failing to recover even the symmetric, already–known case) would be decisive evidence against the algebraic route and would redirect the programme toward purely variational, non algebraic, formulations.

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